

Analysis of real-time kinematic positioning of UAV based on BDS-3 PPP-B2b service

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Abstract

High-precision positioning of unmanned aerial vehicles (UAVs) is generally achieved through real-time kinematic (RTK) technology, which rely on ground base stations. This poses challenges for real-time high-precision positioning in areas lacking base stations. BeiDou navigation satellite system (BDS-3) provides publicly available satellite-based real-time precise point positioning (PPP) service through its geostationary orbit (GEO) satellites, providing a novel method for real-time high-precision UAV positioning. In this study, we attempt to propose a forecast epoch correction (FEC) method to improve the accuracy at the step-change epochs of orbit corrections and meet real-time requirements. The recovered orbit accuracy is improved by 8.6 %, 10.5 %, and 9.3 % in the radial, along-track, and cross-track directions compared to using corrections of the nearest epoch. The three-dimensional (3D) orbit accuracy of PPP-B2b product is improved by 16.7 % compared to the broadcast orbits. We investigated the static PPP accuracy at stations in China and nearby regions, as well as the kinematic PPP accuracy for UAVs. For static PPP, using PPP-B2b service achieved an accuracy of 0.04 m horizontally and 0.08 m vertically within 25 min. The orbits and clock offsets of PPP-B2b product were recovered at 30 s and 5 s intervals for the kinematic UAV positioning. The result shows that it could achieve convergence within 17 min in hovering, forward flight, or mixed flight state. The average accuracy of PPP-B2b products is about 0.064 m, 0.047 m, and 0.123 m in the E, N, and U directions, respectively. These results are slightly worse than those of the Wuhan University Multi-GNSS (WUM) final products, and comparable to the results using the real-time service provided by the Center National 'Etudes Spatiales (CNES) products. The results of 5 s intervals were comparable as those of the 30 s intervals. In sum, the PPP-B2b service can meet the real-time positioning accuracy requirements for UAV missions.

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Keywords: BDS-3; PPP-B2b; UAV; Precise point positioning

1. Introduction

With the development of the global navigation satellite system (GNSS) positioning technology, there is a significant increase in the demand for real-time high-precision

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positioning. The two dominant real-time high-precision positioning technologies are real-time kinematic (RTK) and precise point positioning (PPP), both of which have been widely used in the fields of land, sea and air (Zumberge et al., 1997; Rizos, 2009; Zhang et al., 2017; Li et al., 2022a). Real-time high-precision positioning facilitates the widespread applied of unmanned aerial vehicles (UAVs) in these areas (Guan et al., 2021; Erol et al., 2024). Generally, the position of UAVs relies on RTK technology that is subject to the base stations (Rizos, 2009). The service range of RTK is limited to the communication with the base stations (El-Mowafy, 2012). In contrast, PPP can achieve global high-precision positioning with a single receiver, which means that it gets rid of the limiting requirement of reference station (Zumberge et al., 1997; Li et al., 2015). However, this service requires that users be able to receive the real time high-precision satellite orbit and clock offset products through some means, such as the web or satellite broadcasts. The most commonly used is the web-based real-time PPP service provided by the International GNSS service (IGS) (Elsobeiey and Al-Harbi, 2016). However, poor network connectivity in harsh environments can result in lost real-time corrections, degrading PPP performance (Li et al., 2022b).

Unlike the real-time PPP service from the IGS, the BeiDou navigation satellite system (BDS-3), officially operated on 31 July 2020, provides real-time PPP service through its unique geosynchronous earth orbit (GEO) satellites (Yang et al., 2019). By the end of 2023, the system comprises 4 GEO satellites, 3 inclined geosynchronous orbit (IGSO) satellites, and 26 medium earth orbit (MEO) satellites. The BDS-3 system broadcasts the new B1C (1575.420 MHz), B2b (1207.140 MHz), and B2a (1176.45 MHz) signals based on retaining the B1I and B3I signals (Yang et al., 2020), in which, the B2b signal is specifically designed for PPP enhancement service (PPP-B2b) (Yang et al., 2021). Notably, the PPP-B2b service allows users to receive real-time corrections broadcast by GEO satellites across China and nearby regions, independent of internet connectivity (Liu et al., 2022a). It is possible for UAVs to achieve real-time high-precision positioning through PPP-B2b services in areas with poor connectivity. This PPP-B2b service covers the Asia-Pacific region (80° and 155°E, 5° and 55°N) (Yang et al., 2021), with the best coverage for eastern China (Tao et al., 2021). Note that only the corrections for satellites over in this region are broadcast to users. Likewise, the Galileo and Quasi-Zenith Satellite System (QZSS) also provide their own satellite-based PPP services, which is the so-called High Accuracy Service (HAS) and Centimeter Level Augmentation Service (CLAS) (Martini et al., 2024; Kan et al., 2024).

The BDS-3 PPP-B2b products are currently designed to supports BDS-3 and GPS. In terms of the accuracy of PPP-B2b products, the orbit corrections of the BDS-3 MEO satellites in the radial, along-track, and cross-track directions are less than 0.3 m, with the clock corrections within

2 m (Xu et al., 2024). The root mean square (RMS) error of the radial recovered orbits is about 0.05 m, resulting in a three-dimensional (3D) orbit error of about 0.2 m. The clock offset error is 0.26 ns, and the signal-in-space ranging error (SISRE) reaches 0.8 m (Tang et al., 2022). The positioning performance of PPP-B2b has been evaluated through comparisons with the Center National d' Etudes Spatiales (CNES), HAS, and CLAS real-time services using static stations. Results demonstrate that the performance is comparable to these services, achieving a 3D accuracy of 0.1 m in static PPP and 0.3 m in kinematic PPP (Tao et al., 2021; Kan et al., 2024). A Helmert coordinate transformation-based combination of HAS and PPP-B2b products is proposed, enhancing the applicability of PPP-B2b products (Wei et al., 2025). Apart from simulated kinematic tests, the vehicle-based PPP achieved 0.2 m in both horizontal and vertical directions in the complex urban canyons (Xiao et al., 2023; Zhang et al., 2022). The ship-based real-time experiments realized the overall positioning accuracy of 0.2 m for BDS-3-only on the ocean (Geng et al., 2022). Furthermore, the PPP-B2b service has also been validated for applications in precise orbit determination of low earth orbit (LEO) satellites, time transfer, real-time inversion of precipitable water vapor (PWV), and earthquake monitoring (Shi et al., 2025; Ge et al., 2023; Yang et al., 2023; Zang et al., 2024). The above analyses have demonstrated the accuracy and basic positioning performance of PPP-B2b products. However, there are fewer studies on the kinematic UAV positioning using PPP-B2b service. The kinematic positioning of UAV based on PPP-B2b service is necessary to be carried out to expand the application of UAV without ground station.

Critical factors must be carefully addressed to recover precise orbits and clock offsets for enhancing real-time positioning accuracy, such as satellite-specific clock offset bias, missing corrections, and mismatching problems (Liu et al., 2022b; Sun et al., 2023; Ouyang et al., 2023). The positioning accuracy and convergence speed of static PPP can be improved by 40–50 % by correcting this satellite-specific clock offset bias of PPP-B2b products (Sun et al., 2023; Yuan et al., 2025). The missing corrections can be compensated for by the last updated set of corrections prior to the outage epoch (Chen et al., 2023). Regarding the matching strategy, orbit and clock correction matching should follow the same issue of data for correction (IOD CORR). However, orbit and clock corrections are updated at 48 s and 6 s intervals, respectively. This difference leads to mismatches where rapidly updated clock corrections cannot be matched with the slower-updating orbit corrections during IOD transitions. The nearest available corrections can be utilized to compensate for mismatching corrections (Zhao and Zhai, 2024). Moreover, the broadcast orbit cannot be directly matched with the corresponding orbit correction if users try to solve the position at 5 s or 30 s intervals. A common approach employs the nearest time correction for orbit recovery, which leads to inaccuracies in the recovered orbits. In addition, hourly

step-changes were found in the orbit corrections (Chen et al., 2023), which is consistent with the discontinuity of broadcast ephemeris updates (Liu et al., 2022a). This may potentially increase the recovery errors. Thus, a suitable method for orbit correction matching needs to be proposed to improve the accuracy of the PPP-B2b orbits.

This study is devoted to investigating the kinematic positioning performance of UAVs based on the unique PPP-B2b service without ground support. First, the orbit and clock corrections of the PPP-B2b service were evaluated. The orbit recovery was performed using the common nearest epoch corrections (NEC) method and the new forecast epoch corrections (FEC) method. The orbits and clock offsets of the PPP-B2b service were assessed by the final Wuhan University Multi-GNSS (WUM) products, which were employed as references. Then, the BDS-3-only static PPP performance was analyzed by MGEX stations distributed in China and nearby regions. Finally, the real-time kinematic PPP performance of PPP-B2b is evaluated through kinematic tests based on different flight states of UAVs, and the accuracy are compared with those of WUM and CNES products.

2. Methods

This section provided a brief overview of PPP-B2b products and matching strategies for orbits and clock offsets. Only the BDS-3 system of PPP-B2b products was employed in this study. Two methods for recovering PPP-B2b orbits were discussed due to the characteristics of orbit corrections. Then, the methods of assessment are given for evaluating the recovered orbits and clock offsets. Finally, we introduced the static and kinematic PPP strategies for fixed stations and a mapping UAV.

2.1. Recovery method for orbits and clock offsets

The PPP-B2b products broadcasts 27 BDS-3 satellites (C19–C46) corrections via the B2b signal. Four kinds of correction messages are provided to the users: satellite masks, orbit, clock, and differential code bias (DCB) corrections. These messages are matched with each other according to the issue of data (IOD) (CSNO, 2020). The details of PPP-B2b message types and the IOD parameters are listed in Table 1. In general, the IOD issue of data for state space representation (SSR) is the same in all message types so that it can be used for matching with other message

types at the same epoch. IOD navigation (IODN) is only broadcast with orbit corrections to match the IOD of clock (IODC) from the navigation message. To recover the orbits and clock offsets, users should first collect the civil navigation (CNAV1) message of the B1C signal, and then match its IODC with the IODN to determine which sets of orbit corrections should be used. The clock corrections must ensure that IOD Corr are the same to be used with the corresponding orbit corrections.

The orbit and clock offset of WUM products are at 5 min and 30 s intervals respectively. Those of the PPP-B2b products were recovered at 30 s intervals to facilitate evaluation in this study. The schematic of matching orbit corrections is shown in Fig. 1, with an orbit correction interval of 48 s. The users cannot directly obtain corrections with an interval of 30 s for orbit recovery. The first common method is the NEC. The satellite position is recovered using the orbit corrections closest to the desired epoch T_d . For example, the satellite position at epoch T_d is calculated using the broadcast orbit parameters, and then the correction of the nearest epoch T_{i-1} or T_i is applied for recovery. Furthermore, there is a case where the recovery epoch T_d falls at the epoch when a step-change occurs, i.e., between epoch T_{i-1} and T_i . The NEC method may match corrections with large errors. To address the resulting errors, we propose the so-called FEC method. First, the orbit is recovered at the same epoch of orbit correction, and then the orbit at the user's desired epoch is forecast. For instance, the satellite position at the orbit correction epochs (T_{i-8} , T_{i-7} , T_{i-6} , T_{i-5} , T_{i-4} , T_{i-3} , T_{i-2} , T_{i-1} , T_i) is recovered firstly, and then the position at epoch T_d is forecast through eighth-order Neville's Algorithm. The method can be applied in real-time applications.

The orbit corrections provided by PPP-B2b are divided into the radial (R), along-track (A), and cross-track (C) directions. The satellite position and velocity calculated from the B1C-CNAV1 broadcast ephemeris can be corrected as:

$$X_{orbit} = X_{brdc} - [e_R e_C e_A] \cdot \delta O \quad (1)$$

where X_{orbit} is the corrected satellite position; X_{brdc} is the satellite position calculated from the broadcast orbits; δO refers to the orbit correction vector composed of the radial, along-track, and cross-track directions.

$$e_R = \frac{X_{brdc}}{|X_{brdc}|}$$

Table 1
PPP-B2b messages and navigation messages broadcasted by BDS-3 GEO satellites.

Message	Type	IOD parameters			Interval (s)
Satellite mask	Type 1	IOD SSR	IODP		48
Orbit correction	Type 2	IOD SSR	IODN	IOD Corr	48
DCB	Type 3	IOD SSR			48
Clock correction	Type 4	IOD SSR		IOD Corr	6
Navigation message	BDS-3: CNAV1		IODC		3600

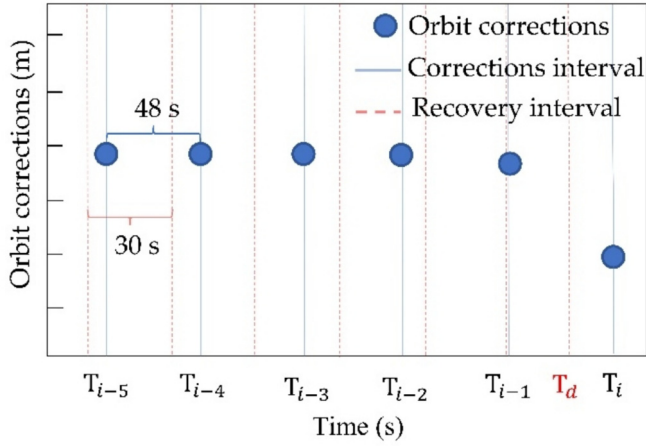


Fig. 1. Matching strategy of orbit corrections. The blue solid denotes the orbit corrections at 48 s. The red dashed line is the epoch to be recovered at 30 s.

$$e_C = \frac{X_{brdc} \times \dot{X}_{brdc}}{|X_{brdc} \times \dot{X}_{brdc}|} \quad (2)$$

$$e_A = e_C \times e_R$$

where $\dot{X}_{brdc}$ is the satellite velocity vector; e_R , e_C and e_A are the unit vector of radial, cross-track and along-track directions, respectively. The clock corrections in message type 4 are used to compensate for the broadcast clock offset. The recovery method is as follows:

$$dT_{B3}^j = dT_{brdc} - \frac{dT_c}{C} \quad (3)$$

where dT_{brdc} is the clock offset referenced to the B3I signal, calculated by broadcast clock offset; dT_c is the clock corrections in meters in the message type 4; C is the speed of light; dT_{B3}^j is the recovered clock offset for the satellite.

2.2. Evaluation method for orbit and clock offset

The WUM products were used as references to assess the orbit and clock offset accuracy of PPP-B2b products. It should be noted that the orbit and clock offset reference of the WUM products and the PPP-B2b products is different (Wang et al., 2023). Therefore, the reference points of satellite orbits were converted as the satellite center of mass (COM) for evaluation, using the satellite phase center offset (PCO) values released by IGS20.atx. Regarding the clock offset, we used the DCB provided by PPP-B2b products to convert to the B1I/B3I ionosphere-free combination.

There are systematic biases between different clock offset products due to various time bases for clock offset estimation. These biases can be eliminated using the double-difference method (Guo et al., 2023). For each epoch, the average of the single-difference for all satellites is computed to reduce the systematic biases.

$$\Delta \nabla dT_{PPP-B2b, WUM}^S = (dT_{PPP-B2b}^S - dT_{WUM}^S) - \frac{1}{N} \sum_{i=1}^N (dT_{PPP-B2b}^i - dT_{WUM}^i) \quad (4)$$

where $\Delta \nabla dT^S$ represents the systematic biases between the PPP-B2b and WUM clock offset for satellite S ; $\frac{1}{N} \sum_{i=1}^N (dT_a^i - dT_b^i)$ means the systematic biases at epoch i ; N is the number of satellites at this epoch. It is important to note that the PPP-B2b service provides corrections only for satellites located in areas with GEO satellite signals (Yang et al., 2022). This means that the set of corrected satellites varies over time, with a period of approximately several hours (He et al., 2023). Therefore, the results of the double-difference method need to be corrected to reduce discontinuities, and the specific method can be referenced from Tao et al. (2021).

SISRE is a key performance indicator for assessing the equivalent ranging error due to orbit and clock offset errors (Montenbruck et al., 2018). We use it to evaluate the accuracy of PPP-B2b products. The computation is as follows:

$$\text{SISRE} = \sqrt{(a_1 dR - dclk)^2 + a_2 (dA^2 + dC^2)} \quad (5)$$

where dR , dA and dC are the orbit errors in the radial, along-track, and cross-track directions, respectively; a_1 and a_2 are the SISRE weight coefficients related to the altitude of the satellites, and the coefficients are 0.981 and 0.136 for BDS MEO satellites, respectively (Montenbruck et al., 2018); $dclk$ is the clock offset errors. Furthermore, the orbit-only SISRE (SISRE-orb) can be computed with the neglect of the clock offset errors in Eq. (5).

2.3. PPP strategy

The type of mini_2 receiver produced by Femtomes was used to receive correction messages of the PPP-B2b from the day of year (DOY) 142 to 148, 2024. This receiver can decode signals of all frequencies of BDS-3 and GPS. It also supports BDS-3 satellite-based augmentation system (SBAS) and PPP-B2b services. The period is from the day of year (DOY) 142 to 148, 2024. As shown in Fig. 2, observations at four MGEX stations in the Asia-Pacific region were collected to verify the accuracy of the PPP-B2b products and PPP algorithm, which is the prerequisite for kinematic PPP. A modified real-time kinematic library (RTKLIB) software (version 2.4.3) was used to process BDS-3-only static PPP for each station. IGS weekly solution coordinates were regarded as the true value to evaluate compute the errors in the East (E), North (N), and Up (U) directions. Static convergence time was defined as the time length from the start epoch to the epoch where the horizontal and vertical error is less than 0.1 m for 20 consecutive epochs.

A DJI M350 mapping UAV was used to evaluate the kinematic PPP performance of the PPP-B2b service. This UAV can decode the BDS-3 observations through its

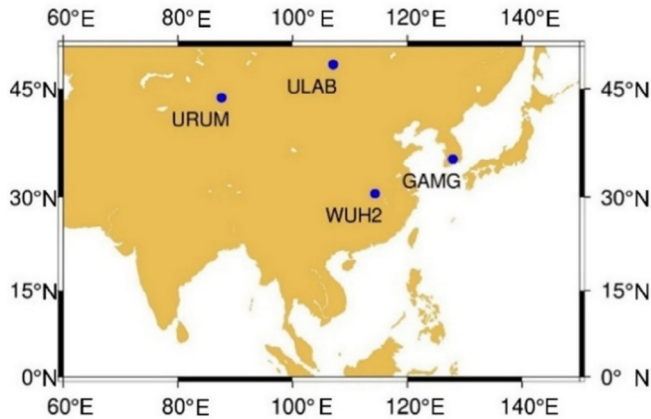


Fig. 2. Distribution of MGEX stations in Asia-Pacific region.

Table 2
Three different flight states for DJI M350 UAV in the kinematic PPP processing on DOY 143 and 144, 2024.

Plan	Flight time	Flight states
FF	07:25-07:55, DOY 144	Forward flight
HF	07:40-08:10, DOY 143	Hovering flight
MF	08:55-09:25, DOY 143	Mixed flight

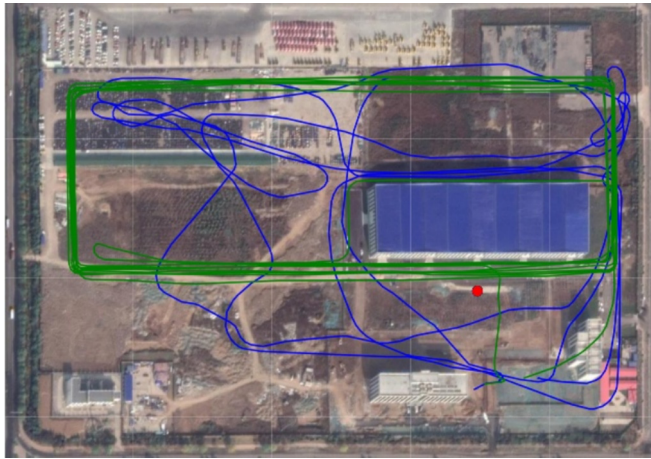


Fig. 3. Flight trajectories of FF (green), HF (red), and MF (blue) for DJI M350 UAV on DOY 143 and 144, 2024.

own receiving equipment, with an average flight time of about half an hour. We collected three groups of observations on 22 to 23 May 2024 at 1 s intervals for three flight states, and the detailed information is shown in Table 2. The forward flight (FF) maintained a regular flight state from 7:25 to 7:55 on DOY 144, 2024. The hovering flight (HF) occurred in a hovering state from 7:40 to 8:10 on DOY 143, 2024. The flight duration of the mixed flight (MF) spanned from 8:55 to 9:25 on DOY 143, 2024. The period between 8:55 and 9:04 involved hovering, followed by a forward flight state from 9:04 to 9:25. The flight trajectories are plotted in Fig. 3. The flight environment of the three experiments was the same, as shown in Fig. 4. The field of view was wide and unobstructed. Observation collected from the DJI M350 needs to be converted into RTCM 3 format. The kinematic PPP mode was conducted using both PPP-B2b and WUM products. The satellite coordinates (X, Y, Z) and receiver clock offset for each epoch are treated as white noise, and the standard single point positioning (SPP) results are used as the initial values for PPP. The specific processing strategy is detailed in Table 3. In addition, a nearby continuously operating reference station (CORS) was used for realizing RTK positioning to provide the UAV's reference path. Kinematic convergence time was defined as the time length from the start epoch to the epoch where the horizontal and vertical error is less than 0.2 m for 20 consecutive epochs.

3. Results

To verify the improved PPP-B2b orbit recovery method, the characteristics of PPP-B2b orbit and clock corrections for BDS-3 MEO satellites were analyzed. The accuracy of the two recovery methods was compared to verify the improved PPP-B2b orbit recovery method. The BDS-3-only static PPP performance was evaluated by four MGEX stations in China and nearby regions. Finally, UAV observations were collected in three different flight states: hovering, forward, and mixed flight. The kinematic PPP performance was tested using these three sets of observation.



Fig. 4. The UAV flight environment in the air (a) and on the ground (b).

3.1. Comparison of orbit recovery methods

The time series of the number of corrected satellites, clock corrections, and orbit corrections provided by the PPP-B2b service on DOY 144 is shown in Fig. 5. C35 was excluded from evaluation and PPP testing due to the lack of PPP-B2b corrections throughout the day. The

Table 3
Processing strategies for PPP using WUM and PPP-B2b products.

Items	Processing strategies
Observation type	BDS-3 B1I/B3I combination
Processing interval	30 s for MGEX stations; 30 and 5 s for UAV
Elevation mask	7°
Ionospheric delay	Ionosphere-free combination
Tropospheric delay	Saastamoinen model (Saastamoinen, 1972) + random walk estimation of zenith wet delay
Antenna PCO	IGS20.atx
Solid tide, ocean tide	IERS Conventions 2010 (Petit and Luzum, 2013)
Ambiguity	Float

PPP-B2b products can averagely enhance 8 to 12 BDS-3 satellites per epoch. The orbit corrections for BDS-3 MEO satellites are less than 0.1 m, 0.2 m, and 0.2 m in the radial, along-track, and cross-track directions, respectively, as the broadcast orbits have been improved by inter-satellite link (ISL) technology (Lv et al., 2020). In contrast, the clock corrections reach about 2 m. Besides, some anomalous clock corrections are observed in Fig. 5. The IODC of such anomalies differs from that of the following epochs. They can also lead to a mismatch with orbit corrections. In this paper, these anomalies were neglected and the next set of clock corrections with identical IODC was used to match the orbit corrections.

Fig. 5 also shows that various step-changes occur approximately every hour in orbit corrections. This phenomenon is attributed to the hourly update of the BDS-3 broadcast ephemeris (Lu et al., 2020), and a set of orbit corrections with the same IODN needs to match broadcast orbits with this epoch. Satellites C21 and C28 were selected to illustrate the orbit errors of the NEC and FEC methods

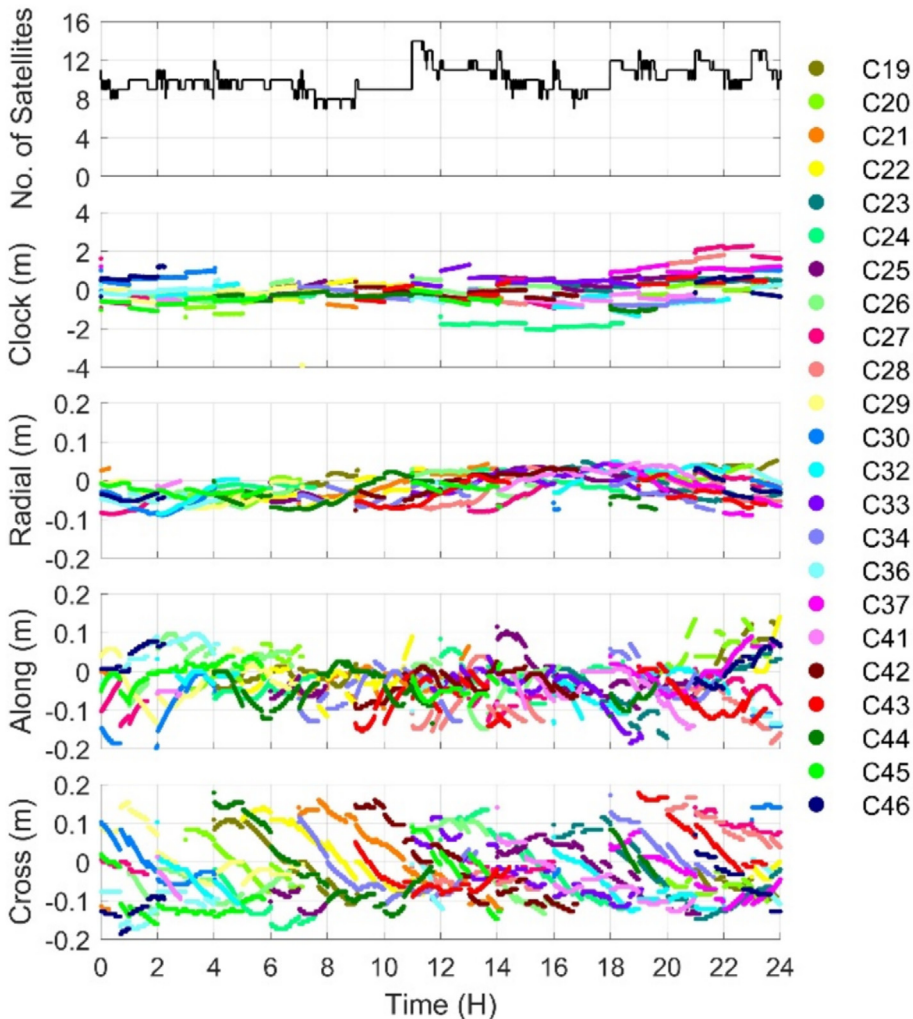


Fig. 5. Time series of the number of recovered satellites, clock corrections, and orbit corrections in radial, along-track, and cross-track directions for BDS-3 MEO satellites provided by PPP-B2b on DOY 144, 2024.

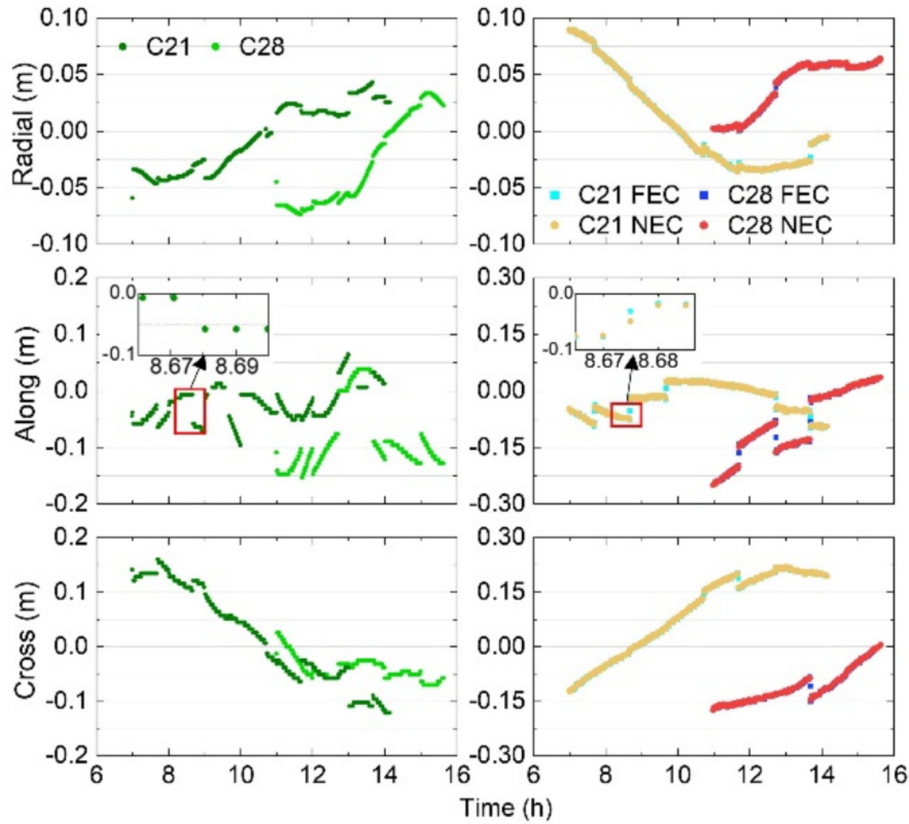


Fig. 6. Time series of orbit corrections of PPP-B2b products (left) and recovered orbit errors using FEC and NEC methods (right).

around the epochs of step-change as shown in Fig. 6. It can be seen that the discontinuity of orbit corrections is more pronounced in the along-track direction compared to the other directions. The orbit error obtained using the FEC method is smaller than that of the NEC method at the step-change epochs. At other epochs, the orbit errors obtained by the FEC method are also comparable to those

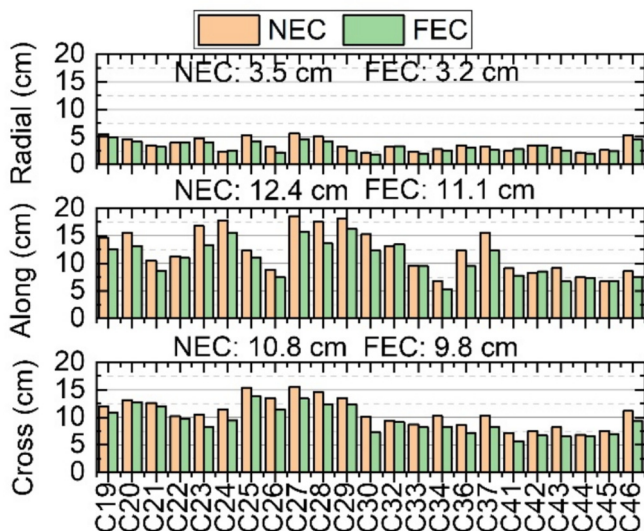


Fig. 7. Recovered orbit errors for the BDS-3 MEO satellites using the FEC and NEC methods at the step-change epochs.

of the NEC method. Fig. 7 compares the recovered orbit errors computed by two recovery methods at these epochs. The average errors of the FEC method are 3.2 cm, 11.1 cm, and 9.8 cm in the radial, along-track, and cross-track directions, respectively, with an improvement of 8.6 %, 10.5 %, and 9.3 % over the NEC method. However, the NEC method can correct more epochs, as the FEC method is unable to recover the orbits at the beginning and end of an orbit arc due to insufficient data. Consequently, a combination of both methods was employed in the subsequent analyses. The NEC method was employed for the epochs at the beginning and end of the step-change, while the FEC method was applied to the remaining epochs. Table 4 summarizes the errors using the two recovery methods and the combination of both methods for BDS-3 MEO satellites. The accuracy of the combined FEC and NEC method shows improvements of 4.5 %, 4.4 %, and 2.2 % in the radial, along-track, and cross-track directions, respectively, compared to the NEC method. The accuracy of the combined FEC and NEC method is comparable to that of the FEC method, with no loss of epochs during the recovery process. It can be applied to real-time services.

3.2. Accuracy of recovered orbit and clock offset

The orbit errors of BDS-3 MEO satellites in the radial, along-track, and cross-track directions on DOY 144, 2024 are plotted in Fig. 8. The errors in the radial direction of

Table 4
Recovered orbit errors for the BDS-3 MEO satellites using the FEC, NEC, and the combined FEC and NEC methods.

Method	Radial (cm)	Along (cm)	Cross (cm)	3D (cm)	Number of recovered epochs
FEC	4.1	14.8	13.8	20.7	970
NEC	4.4	15.6	14.2	21.5	991
Combined FEC and NEC	4.2	14.9	13.9	20.8	991

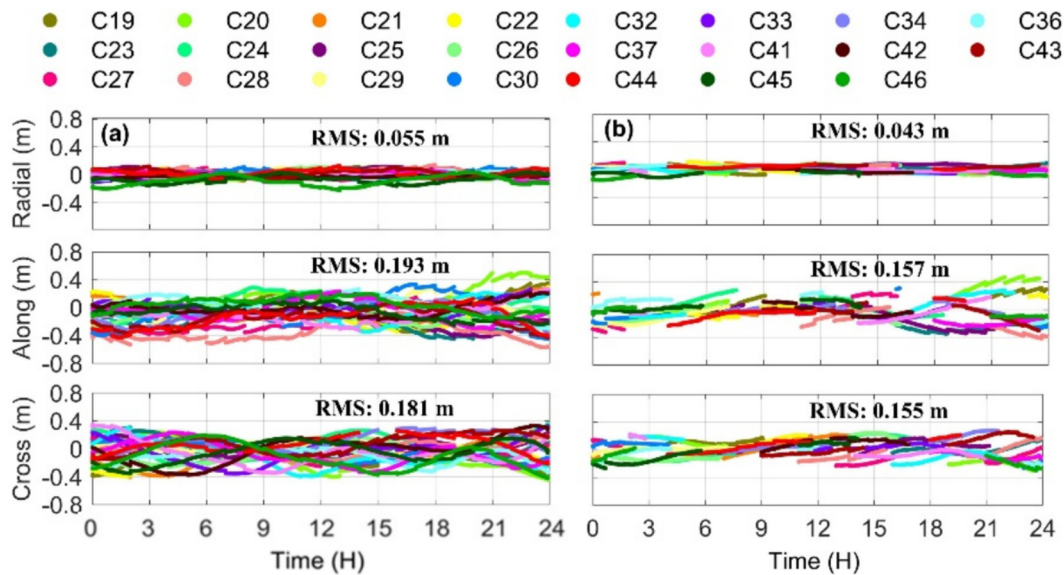


Fig. 8. Errors of the broadcast orbits (a) and PPP-B2b orbits (b) for BDS-3 MEO satellites with respect to the WUM products on DOY 144, 2024.

the broadcast and PPP-B2b orbits are smaller than those in the along-track and cross-track directions, with most radial errors within 0.1 m. The broadcast orbit errors exhibit hourly discontinuities due to the hourly updates of the BDS-3 broadcast ephemeris. After applying the PPP-B2b corrections, the orbit error sequences become much more continuous and smoother. The RMS of orbit errors in the three directions from DOY 142 to DOY 148 in 2024 were also summarized in Fig. 8. The PPP-B2b orbits exhibited errors of 0.043 m, 0.157 m, and 0.155 m in the radial, along-track and cross-track directions, respectively, which are improved by 21.8 %, 18.6 %, and 14.4 % compared to the broadcast orbits.

The results of the double-difference method for clock offset evaluation are depicted in Fig. 9. The errors of broadcast clock offsets exhibited obvious discontinuities when the IOD of navigation messages was updated (Ren et al., 2021). In contrast, the clock offset errors of PPP-B2b following the method in Section 2.3 appear smoother and more continuous. However, the errors show an overall systematic bias compared to broadcast clock offsets. Generally, the stability of the clock offset has an impact on the accuracy of PPP. Therefore, the average STD value of the PPP-B2b clock offsets is about 0.18 ns, which is

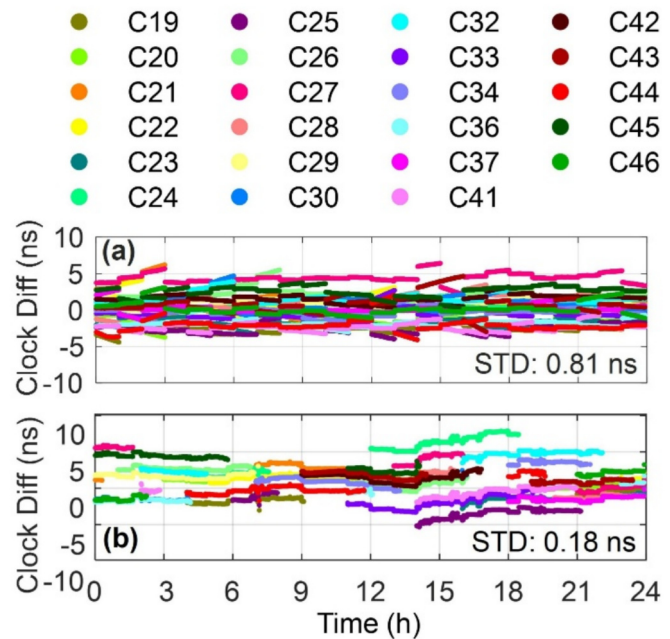


Fig. 9. Series of errors for the broadcast (a) and PPP-B2b (b) clock offset of the BDS-3 MEO satellites with respect to the WUM clock offsets on DOY 144, 2024.

significantly better than the 0.81 ns of the broadcast clock offset.

The RMS of SISRE and SISRE-orb of broadcast and PPP-B2b products are depicted in Fig. 10. In terms of SISRE-orb, the radial component error is the main source of error (Ren et al., 2021). Thus, the RMS for each satellite in the PPP-B2b products is slightly lower than that in the

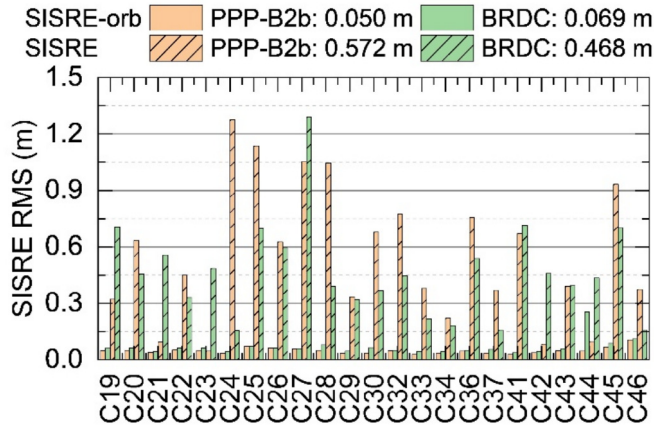


Fig. 10. RMS of SISRE and SISRE-orb for BDS-3 MEO satellites in PPP-B2b and broadcast products from DOY 142 to 148, 2024.

broadcast products, with an average RMS of 0.050 m compared to 0.069 m for the broadcast products. However, the average RMS of the SISRE for PPP-B2b is 0.572 m, which is higher than the 0.468 m of the broadcast ephemeris. This is due to the systematic biases in clock offsets of PPP-B2b (Xue et al., 2021), leading to a larger RMS of SISRE (Sun et al., 2023).

3.3. Static PPP performance for MGEX stations

The BDS-3-only static PPP was conducted using both the PPP-B2b and the WUM products for the ULAB, URUM, GAMG, and WUH2 stations from DOY 142 to 148, 2024. The processing was conducted every 3 h and the results of DOY 144 were plotted in Fig. 11. It can be seen that both products can converge quickly to less than 0.1 m. In the case of WUM products, the positioning accuracy becomes stable after convergence, with no tendency for divergence. For the PPP-B2b products, the error curve is not as smooth as that of WUM products after convergence. The positioning accuracy is calculated from the convergence epoch, and the accuracy and convergence time of 7 days were summarized in Table 5. The accuracy converged to less than 10 cm along the E, N, and U directions within 23 to 28 min. The convergence speed is slightly

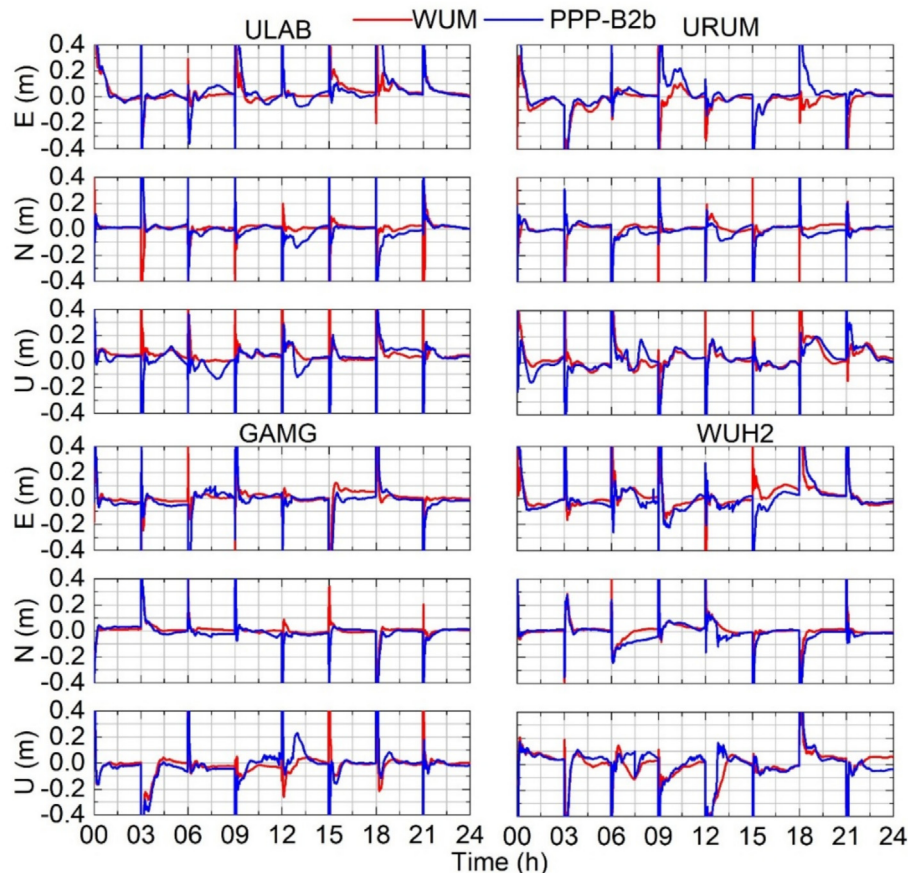


Fig. 11. Error series of static PPP conducted every 3 h using WUM and PPP-B2b products for four stations in China and nearby regions on DOY 144, 2024.

Table 5

Statistical results of accuracy and convergence time for the static PPP performance tests conducted every 3 h using WUM and PPP-B2b products from DOY 142 to 148 in 2024.

Stations	Precise products	E (cm)	N (cm)	U (cm)	3D (cm)	Convergence time (min)
ULAB	WUM	3.6	2.1	5.8	7.1	23
	PPP-B2b	4.1	3.0	7.6	9.1	26
URUM	WUM	3.5	2.5	6.1	7.5	24
	PPP-B2b	4.7	3.1	7.5	9.4	28
WUH2	WUM	4.3	2.0	5.5	7.3	25
	PPP-B2b	5.3	2.5	7.5	9.5	26
GAMG	WUM	3.0	1.3	5.2	6.1	24
	PPP-B2b	5.1	2.5	7.6	9.5	27
Mean	WUM	3.6	2.0	5.7	7.0	24
	PPP-B2b	4.8	2.8	7.6	9.4	27

slower than that of WUM products. The average accuracy is 0.048 m, 0.028 m, and 0.076 m in the E, N, and U directions, respectively. In contrast, the WUM products had an average accuracy of 0.036 m, 0.020 m, and 0.057 m in the E, N, and U directions, respectively. The convergence time is similar for PPP-B2b and WUM products.

3.4. Kinematic PPP performance for UAV

For UAV positioning, the kinematic flight of a UAV may affect positioning accuracy or convergence time. The kinematic PPP performance of UAVs was tested with different states listed in Table 2. The precise positioning of FF, HF, and MF were estimated using PPP-B2b, CNES, and WUM products. The reference flight paths were estimated by the RTK function of RTKLIB. The positioning errors and 3D flight speed are plotted in Fig. 12.

In the FF experiment, the UAV maintained a flight speed of approximately 6 m/s. The results indicate that the positioning errors curves of three products are smoother in the E and N directions compared to the U direction. To evaluate the impact of flight state and speed on convergence time and positioning accuracy, two additional experiments (HF and MF) were conducted. Regarding the HF experiment, the errors indicate fluctuations in the E and U directions during the first 10 min, despite the UAV maintaining a hovering state. In terms of the MF experiment, the UAV was in a hovering state from 8:55 to 9:03, followed by kinematic flight at 4–6 m/s from 9:03 to 9:25. However, it turns out that the positioning error has not been improved during the hovering period. There were fluctuations in the E and U directions. The positioning accuracy remains stable during the kinematic flight after convergence. The results of three experiments show that the flight speed and state have no great influence on the positioning accuracy.

We also found that the positioning accuracy using PPP-B2b, CNES, and WUM products can converge to less than 1 m in the E, N, and U directions within 5 min among three experiments. The most pronounced differences were

observed during the initial 10 min, which can reach 10–50 cm in the U direction. These differences were not significantly related to flight state. During the first 20 min of each experiment, the results of WUM and CNES products exhibit superior positioning accuracy compared to those of PPP-b2b products. The largest differences occurring in the U direction and a smaller one in the N direction. The PPP-B2b products converged more slowly in the U direction compared to CNES products.

The accuracy and convergence of the FF, HF and MF experiments are summarized in Table 6. It can be seen that the convergence speed of HF and MF experiments are about 35 % faster than the FF experiment. This may be due to the fact that the slightly faster convergence speed of positioning accuracy during hovering. In the FF, HF and MF experiments, the average convergence is achieved within 16 to 17 min. It means that the UAV attained positioning accuracies of 0.2 m and 0.4 m in the horizontal and vertical directions, respectively, within approximately 16 min. In terms of the positioning accuracy, the average RMS of positioning accuracy in the E, N, and U directions using the CNES products were 0.063 m, 0.047 m, and 0.114 m, respectively. In contrast, the PPP-B2b service showed comparable performance with 0.064 m, 0.047 m, and 0.123 m respectively. The positioning accuracy of PPP-B2b in the E, N, and U directions is 16.4 %, 20.5 %, and 11.8 % lower than that of WUM products, respectively. Nevertheless, it demonstrated that the PPP-B2b service can meet the real-time positioning accuracy requirements for UAV missions. In contrast, we also recovered the PPP-B2b product to 5 s intervals for kinematic UAV positioning. The accuracy and convergence time are compared in Table 7. It can be seen that the positioning accuracy and convergence speed of the 5 s intervals are slightly improved, especially in the FF and HF experiments.

Furthermore, the error curves of PPP-B2b exhibited instability in the E and N directions, showing discrepancies compared to WUM results during the FF experiment from 07:41 to 07:45. The number of available satellites and the

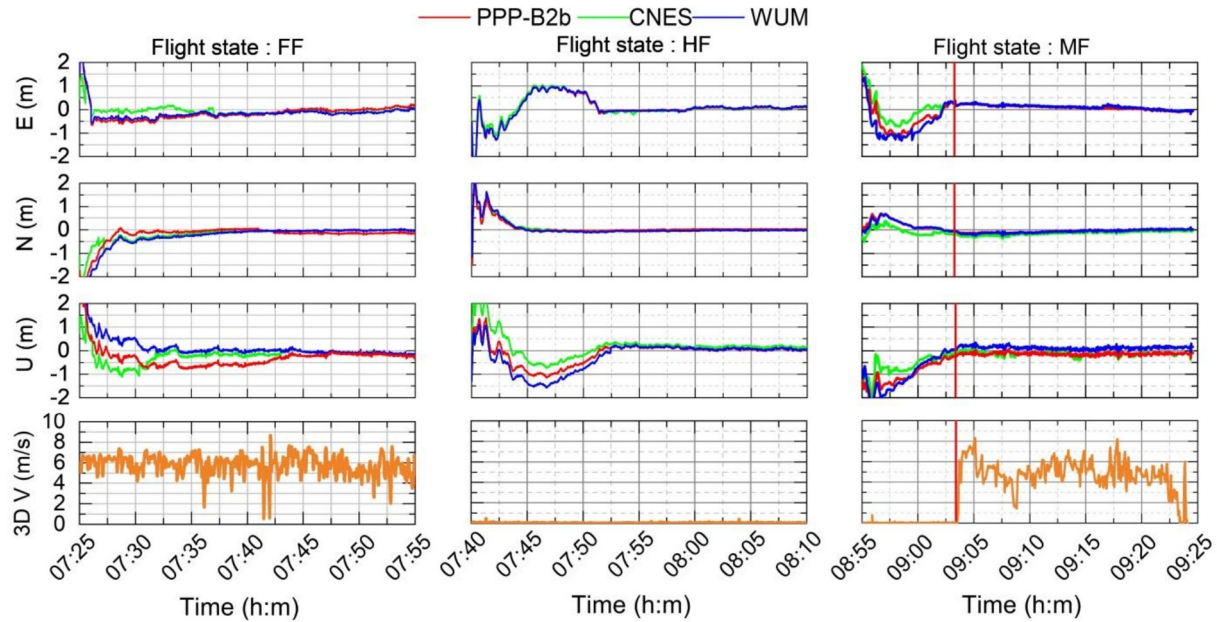


Fig. 12. Kinematic PPP errors and flight speed of FF (left), HF (middle) and MF (right) experiments for the UAV using PPP-B2b, CNES and WUM products.

Table 6

Three sets for kinematic PPP results of accuracy and convergence time using WUM and PPP-B2b products at 30 s intervals.

Plan	Precise products	E (cm)	N (cm)	U (cm)	3D (cm)	Convergence time (min)
FF	CNES	7.0	5.6	10.8	14.0	20
	PPP-B2b	6.8	5.2	12.3	15.0	20
	WUM	5.1	3.8	10.1	11.9	18
HF	CNES	5.9	3.8	11.8	13.7	14
	PPP-B2b	6.2	4.1	12.5	14.5	17
	WUM	6.0	3.3	11.2	13.1	15
MF	CNES	5.9	4.6	11.5	13.7	12
	PPP-B2b	6.3	4.9	12.1	14.5	13
	WUM	5.5	4.5	11.8	13.8	13
Mean	CNES	6.3	4.7	11.4	13.8	16
	PPP-B2b	6.4	4.7	12.3	14.7	17
	WUM	5.5	3.9	11.0	12.9	16

Table 7

UAV kinematic PPP results for PPP-B2b products at 30 s and 5 s intervals.

Plan	Interval	E (cm)	N (cm)	U (cm)	3D (cm)	Convergence time (min)
FF	30 s	6.8	5.2	12.3	15.0	20
	5 s	6.3	4.6	12.0	14.3	19
HF	30 s	6.2	4.1	12.5	14.5	17
	5 s	6.2	3.8	12.3	14.2	16
MF	30 s	6.3	4.9	12.1	14.5	13
	5 s	6.4	4.8	12.1	14.5	13

position dilution of precision (PDOP) are displayed in Fig. 13. The visible satellites and the PDOP values for the PPP-B2b and WUM products are consistent, with approximately 2.2 and an average of 6 to 7 satellites, respectively, indicating that these factors do not account for the differences in positioning accuracy between the

PPP-B2b and WUM products. To investigate further, Fig. 14 illustrates the orbit corrections of PPP-B2b from 07:20 to 08:00. It can be seen that some satellites showed orbital correction jumps in three directions around 07:41. These jumps likely contribute to the observed decrease in PPP-B2b positioning accuracy at the same time.

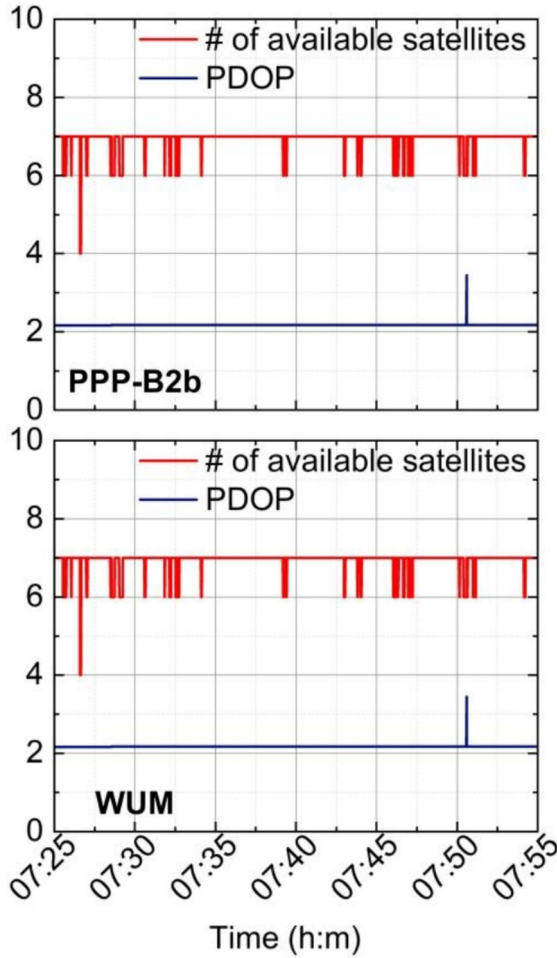


Fig. 13. Number of available BDS-3 satellites and PDOP values during the FF experiment positioning of the UAV on DOY 144, 2024.

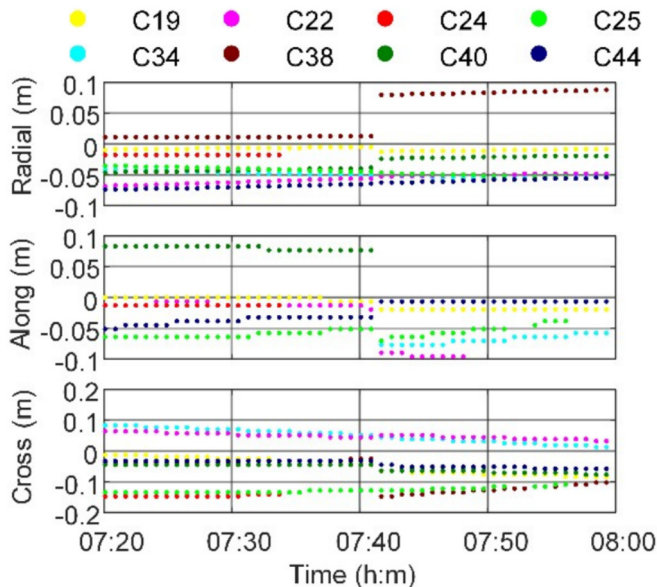


Fig. 14. PPP-B2b orbit corrections of available satellites during the FF experiment positioning of the UAV on DOY 144, 2024.

4. Conclusions

This study investigated the real-time kinematic positioning performance of the UAV based on the PPP-B2b service, with a thorough analysis and assessment of its positioning accuracy. The PPP-B2b correction messages from DOY 142 to 148, 2024 were collected to correct the broadcast orbits and clock offsets. A forecast epoch corrections (FEC) method was proposed to reduce the orbit errors at the step-change epochs, with an improvement of 8.6 %, 10.5 %, and 9.3 % in the radial, along-track, and cross-track directions, respectively. To meet real-time requirements, enhance orbit accuracy and maximize the orbit arcs, we employ the FEC method to forecast corrections. On this basis, the orbit accuracy for step-change epochs was improved, and the NEC method was combined with the FEC method. The combined FEC and NEC method improves the accuracy by 4.5 %, 4.4 %, and 2.2 % in the radial, along-track, and cross-track directions, respectively, compared to using the NEC method.

The characteristics of the PPP-B2b orbit and clock offset corrections are analyzed. The recovered orbits, clock offsets, and the corresponding SISRE were evaluated to verify the proposed PPP-B2b orbit recovery method. The orbits showed an improvement of 21.8 %, 18.6 %, and 14.4 % in the radial, along-track, and cross-track directions, respectively, with respect to the broadcast orbit. The STD of the recovered clock offset was reduced to 0.18 ns, but there is still an inherent systematic bias that results in a SISRE of 0.572 m.

Finally, the accuracy of the PPP-B2b products and algorithm was verified by static PPP experiments. In static positioning, its accuracy reached 0.048 m, 0.028 m, and 0.076 m in the E, N, and U directions within 25 min, which is slightly worse than results using WUM products. The PPP-B2b precise products with different intervals (30 s and 5 s) was used to test kinematic UAV positioning with the same interval. In the 30 s intervals experiment, the PPP-B2b average positioning accuracy under hovering, forward and mixed flight states reach 0.064 m, 0.047 m, and 0.123 m in the E, N, and U directions. The value are slightly worse than that of WUM by 16.4 %, 20.5 %, and 11.8 %, respectively. Compared with CNES products, the horizontal positioning accuracy is comparable, and the vertical accuracy is slightly worse. The results for the three sets of UAV kinematic PPP indicate that the hovering state of UAVs may accelerate the convergence speed of kinematic PPP. Furthermore, we found that the positioning results of the 5 s intervals experiment were comparable to those of 30 s intervals. In sum, UAVs can achieve real-time high-precision positioning through BDS-3 PPP-B2b service.

Currently, PPP-B2b services are limited to ambiguity-float solutions due to the absence of phase bias products in the service. The method would be explored for achieving ambiguity-fixed solutions of the PPP-B2b service. More different complex scenarios should be used to conduct UAV kinematic experiments in the future.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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